

Endogenous Network Formation in Competitive Portfolio Games: A Graphon Equilibrium Approach

1. Introduction

Relative performance concerns are a central feature of modern asset management. Portfolio managers are routinely evaluated against peers, benchmark indices, or custom peer groups, and compensation schemes often depend explicitly on relative performance metrics. Such incentives generate strategic interactions among investors, altering risk-taking behavior and equilibrium allocations. Theoretical analyses of competitive portfolio choice date back to (1), (2), (3), and (?), and have been further developed in continuous-time equilibrium settings with heterogeneous agents and delegated management (e.g., (4), (5), (6)).

A growing strand of the literature studies large-population limits of competitive portfolio games using mean-field techniques. Mean field games (MFG), introduced by (7) and (8), provide tractable approximations of strategic interactions in large systems. Recent work has adapted MFG methods to portfolio competition and asset pricing (e.g., (?), (9)). More recently, heterogeneous interaction structures have been incorporated via graphon limits and network mean field games (10), (?). In these models, the interaction kernel governing peer comparisons is exogenously specified.

In parallel, the financial networks literature emphasizes that interconnections among institutions play a crucial role in systemic risk propagation (11), (12), (?). However, most network models treat links as primitive or institutionally fixed, rather than chosen as part of an equilibrium.

The present paper bridges these strands by endogenizing the interaction network in a continuous-time competitive portfolio game. Rather than solving equilibrium *on* a given graphon, we characterize equilibrium *of* the graphon itself.

Specifically, we consider a continuum of agents with CARA preferences who trade a risky asset and evaluate terminal wealth relative to a weighted average of peers. Unlike existing graphon-based portfolio games, each agent chooses not only a dynamic trading strategy but also a measurable network kernel that determines the weights assigned to other agents' performance. Network formation is costly and convex, which ensures well-posedness and interior equilibria under suitable conditions.

The equilibrium concept is joint: a kernel G^* and portfolio profile π^* form an equilibrium if (i) π^* solves the competitive portfolio game given G^* , and (ii) each agent's

kernel row $G^*(u, \cdot)$ maximizes her equilibrium certainty equivalent net of link costs. The equilibrium portfolios satisfy a resolvent equation

$$\pi^* = (I - \rho \mathcal{G}^*)^{-1} a,$$

where $\mathcal{G}^*$ is the endogenous graphon operator. This representation reveals a spectral amplification mechanism: the leading eigenvalue of $\mathcal{G}^*$ governs systemic risk amplification, and a critical competition threshold emerges at $\rho = 1/\lambda_{\max}$.

Our contributions are fourfold.

First, we prove existence of joint network–portfolio equilibria in infinite-dimensional strategy spaces under quadratic link costs and moderate competition intensity. The proof combines operator-theoretic arguments with fixed-point methods in reflexive Banach spaces.

Second, we derive a spectral characterization of equilibrium amplification. The endogenous network determines the resolvent multiplier $(I - \rho \mathcal{G}^*)^{-1}$, which links network centrality to aggregate risk exposure. A phase transition occurs when competition intensity approaches the inverse spectral radius of the equilibrium graphon.

Third, we characterize the topology of endogenous networks. Under heterogeneous risk tolerance, equilibrium kernels exhibit assortative clustering, with risk-tolerant agents forming more intense connections. As competition intensifies, network centralization increases and converges toward a dominant eigenmode.

Fourth, we establish that decentralized network formation is generically inefficient. Agents internalize private amplification benefits but ignore systemic externalities induced through equilibrium portfolios. Only under homogeneity does decentralized formation coincide with the planner’s allocation.

The model extends existing graphon portfolio games by internalizing network architecture, and connects competitive benchmarking to systemic risk through a tractable operator-theoretic mechanism. More broadly, it contributes to the growing literature on endogenous network formation in economic environments (e.g., (13), (?)), by embedding network choice within a continuous-time financial equilibrium.

The results suggest that competitive pressures do not merely distort portfolio allocations; they reshape the topology of financial networks. Understanding this endogenous structure is essential for assessing systemic vulnerability in large asset management systems.

1.1. Relation to Exogenous Graphon Portfolio Games

Our framework is closely related to the graphon portfolio game studied in (14), which analyzes competitive portfolio choice in large populations with heterogeneous interaction structures.

Tangpi and Zhou (14) consider a sequence of finite-player games with random heterogeneous benchmarking weights. They establish existence of Nash equilibria for the finite system and prove convergence to an infinite-population limit characterized by a

graphon-based McKean–Vlasov forward–backward SDE. Their primary contribution is analytical: they show backward propagation of chaos and well-posedness of quadratic graphon FBSDE systems. In their model, the interaction kernel is exogenous, arising as the limit of random adjacency matrices.

The present paper departs from this structure along a fundamentally different dimension. Rather than taking the graphon as primitive, we treat the interaction kernel as an equilibrium object. Agents choose whom to benchmark against, subject to convex link costs, and the resulting graphon emerges jointly with equilibrium portfolios.

This modification alters the mathematical structure in a nontrivial way. In (14), the infinite-dimensional system is a fixed graphon FBSDE. In contrast, our model introduces a fixed-point problem over the space of measurable kernels. The graphon operator itself becomes endogenous and enters the resolvent equation governing equilibrium portfolios. As a consequence, spectral properties of the equilibrium kernel determine amplification and stability thresholds.

Conceptually, the distinction is between equilibrium *on* a network and equilibrium *of* a network. The exogenous graphon framework in (14) analyzes how heterogeneous benchmarking shapes portfolio choice. We analyze how competitive incentives reshape the benchmarking structure itself.

The two approaches are complementary. Exogenous graphon models are natural when interaction structures are institutionally fixed. Endogenous graphon formation is relevant when agents can strategically select peer groups, benchmark indices, or performance comparison sets.

Our results show that once network formation is internalized, competition generates centralization, assortative clustering, and spectral amplification effects that do not arise when the kernel is fixed.

2. Model

2.1. Probabilistic Framework

Fix a finite horizon $T > 0$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space supporting an m -dimensional Brownian motion $W = (W_t)_{t \in [0, T]}$. Let $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ be the augmented filtration generated by W .

Agents are indexed by $u \in I := [0, 1]$. The unit interval is equipped with its Borel σ -algebra and Lebesgue measure. All equalities involving u are understood to hold for almost every $u \in I$.

Throughout, measurability in u and ω is understood with respect to the product space $(I \times \Omega, \mathcal{B}(I) \otimes \mathcal{F})$.

2.2. Preferences and Heterogeneity

Each agent u has CARA preferences with heterogeneous risk aversion $\gamma(u) > 0$, assumed measurable and bounded away from zero and infinity.

Terminal utility takes the form

$$U_u(x) = -\exp(-\gamma(u)x).$$

CARA preferences ensure that equilibrium portfolios are independent of wealth levels, which is crucial for tractability in large populations.

2.3. Financial Market

There is one riskless asset with rate $r = 0$, and a d -dimensional risky asset whose price process follows

$$dS_t = \mu dt + \Sigma dW_t,$$

where:

- $\mu \in \mathbb{R}^d$ is constant,
- $\Sigma \in \mathbb{R}^{d \times m}$ has full rank,
- $M := \Sigma \Sigma^\top$ is positive definite.

Markets are incomplete when $m < d$, which introduces nontrivial risk-sharing structure.

2.4. Admissible Trading Strategies

A strategy for agent u is an $\mathbb{F}$ -predictable process $\pi(u) = (\pi_t(u))_{t \in [0, T]}$ with values in $\mathbb{R}^d$ satisfying

$$\mathbb{E} \left[\int_0^T \|\pi_t(u)\|^2 dt \right] < \infty.$$

The associated wealth process evolves as

$$dX_t(u) = \pi_t(u)^\top \mu dt + \pi_t(u)^\top \Sigma dW_t, \quad X_0(u) = x_0(u).$$

Given square-integrability of π , $X_T(u)$ is Gaussian.

2.5. Relative Performance Benchmarking

Agents care about performance relative to peers.

A measurable kernel

$$G : I \times I \rightarrow \mathbb{R}_+$$

with

$$\int_0^1 G(u, v) dv = 1 \quad \text{for all } u$$

represents benchmarking weights.

Given G , agent u evaluates terminal wealth relative to

$$B_T^G(u) := \int_0^1 G(u, v) X_T(v) dv.$$

The relative-performance-adjusted payoff is

$$Z_T(u) = X_T(u) - \rho B_T^G(u),$$

where $\rho \geq 0$ measures competition intensity.

The agent maximizes

$$\mathbb{E}[-\exp(-\gamma(u)Z_T(u))].$$

2.6. Endogenous Network Formation

Unlike existing graphon portfolio games, the kernel G is not exogenous. Each agent chooses her row

$$G(u, \cdot) \in L^2([0, 1]),$$

subject to:

1. $G(u, v) \geq 0$ for all v ,
2. $\int_0^1 G(u, v) dv = 1$.

We denote by $\mathcal{G}$ the induced integral operator:

$$(\mathcal{G}f)(u) = \int_0^1 G(u, v)f(v) dv.$$

Network formation is costly. We assume a strictly convex quadratic cost:

$$C(G(u, \cdot)) = \frac{\kappa}{2} \int_0^1 G(u, v)^2 dv, \quad \kappa > 0.$$

Quadratic costs ensure strict convexity and interior equilibria.

2.7. Agent's Optimization Problem

Given a population strategy profile $v \mapsto \pi(v)$ and a kernel G , agent u solves

$$\sup_{\pi(u), G(u, \cdot)} \mathbb{E} \left[-\exp \left(-\gamma(u)(X_T(u) - \rho B_T^G(u)) \right) \right] - C(G(u, \cdot)).$$

Because $Z_T(u)$ is Gaussian, Lemma A.1 implies that this problem is equivalent to maximizing a mean–variance functional.

2.8. Equilibrium Definition

A pair (π^*, G^*) is a joint equilibrium if:

1. For fixed G^* , π^* is a Nash equilibrium of the competitive portfolio game.
2. For fixed π^* , each $G^*(u, \cdot)$ maximizes agent u 's objective over admissible kernels.

Thus, equilibrium is taken over an infinite-dimensional strategy space combining dynamic controls and measurable network kernels.

3. Equilibrium Characterization

In this section we characterize the joint equilibrium of the endogenous network–portfolio game. We proceed in three steps: first reducing the utility to a mean–variance form, then solving the portfolio best response for a fixed network, and finally deriving the optimal network given portfolios.

Throughout, G denotes a measurable kernel satisfying

$$G(u, v) \geq 0, \quad \int_0^1 G(u, v) dv = 1,$$

and $\mathcal{G}$ denotes the induced operator on $L^2(I)$:

$$(\mathcal{G}f)(u) = \int_0^1 G(u, v)f(v) dv.$$

3.1. Gaussian Reduction

Under admissible strategies, terminal wealth is Gaussian:

$$X_T(u) = x_0(u) + \int_0^T \pi_t(u)^\top \mu dt + \int_0^T \pi_t(u)^\top \Sigma dW_t.$$

Hence the relative payoff

$$Z_T(u) = X_T(u) - \rho \int_0^1 G(u, v)X_T(v) dv$$

is also Gaussian.

For a Gaussian random variable Z ,

$$\mathbb{E}[-e^{-\gamma Z}] = -\exp\left(-\gamma \mathbb{E}[Z] + \frac{\gamma^2}{2} \text{Var}(Z)\right).$$

Therefore maximizing expected CARA utility is equivalent to maximizing

$$\mathcal{J}(u) = \mathbb{E}[Z_T(u)] - \frac{\gamma(u)}{2} \text{Var}(Z_T(u)) - C(G(u, \cdot)).$$

Thus the stochastic control problem reduces to a static mean–variance functional.

3.2. Portfolio Best Response for Fixed Network

Fix a kernel G . Since the market coefficients are constant and preferences are CARA, the optimal portfolio is time-independent. Write $\pi(u)$ for the constant strategy.

We compute:

$$\mathbb{E}[X_T(u)] = T \pi(u)^\top \mu,$$

$$\text{Var}(X_T(u)) = T \pi(u)^\top M \pi(u), \quad M := \Sigma \Sigma^\top.$$

Using linearity of covariance, one obtains

$$\mathcal{J}(u) = T \pi(u)^\top \mu - \rho T \int_0^1 G(u, v) \pi(v)^\top \mu dv - \frac{\gamma(u) T}{2} \left(\pi(u)^\top M \pi(u) - 2\rho \pi(u)^\top M(\mathcal{G}\pi)(u) + \rho^2 \|(\mathcal{G}\pi)(u)\|_M^2 \right),$$

where

$$\|x\|_M^2 := x^\top M x.$$

Taking the first-order condition with respect to $\pi(u)$ yields

$$\mu - \gamma(u) M \pi(u) + \rho \gamma(u) M(\mathcal{G}\pi)(u) = 0.$$

Multiplying by M^{-1} gives

$$\pi(u) = \frac{1}{\gamma(u)} M^{-1} \mu + \rho(\mathcal{G}\pi)(u).$$

Define

$$a(u) := \frac{1}{\gamma(u)} M^{-1} \mu.$$

In operator form,

$$\pi = a + \rho \mathcal{G}\pi.$$

Thus,

$$(I - \rho \mathcal{G})\pi = a.$$

Whenever $I - \rho \mathcal{G}$ is invertible on $L^2(I)$,

$$\boxed{\pi = (I - \rho \mathcal{G})^{-1} a.}$$

This is the portfolio best response given G .

3.3. Network Best Response for Fixed Portfolio

Fix π . We now optimize $G(u, \cdot)$.

From the mean–variance expression, only the covariance term depends on G . One obtains

$$\mathcal{J}_G(u) = \rho\gamma(u)\pi(u)^\top M(\mathcal{G}\pi)(u) - \frac{\kappa}{2} \int_0^1 G(u, v)^2 dv,$$

subject to normalization.

Define

$$\Lambda(u, v) := \rho\gamma(u)\pi(u)^\top M\pi(v).$$

The problem becomes

$$\max_{G(u, \cdot)} \int_0^1 \Lambda(u, v)G(u, v)dv - \frac{\kappa}{2} \int_0^1 G(u, v)^2 dv$$

subject to

$$\int_0^1 G(u, v)dv = 1.$$

This is a strictly concave quadratic problem. The Lagrangian is

$$L = \int_0^1 \Lambda G - \frac{\kappa}{2} \int_0^1 G^2 + \lambda(u) \left(1 - \int_0^1 G \right).$$

The first-order condition gives

$$\Lambda(u, v) - \kappa G^*(u, v) - \lambda(u) = 0.$$

Hence

$$G^*(u, v) = \frac{1}{\kappa} \Lambda(u, v) + \frac{\lambda(u)}{\kappa}.$$

The multiplier $\lambda(u)$ is determined by normalization.

Thus the optimal kernel is affine in the covariance structure of equilibrium portfolios.

3.4. Joint Equilibrium as Fixed Point

Combining the two best responses, a joint equilibrium (π^*, G^*) satisfies

$$\pi^* = (I - \rho\mathcal{G}^*)^{-1}a,$$

$$G^*(u, \cdot) = \text{BestResponse}(\pi^*).$$

Equivalently, π^* is a fixed point of the mapping

$$\Phi_\rho(\pi) = (I - \rho\mathcal{G}(\pi))^{-1}a.$$

A joint equilibrium is therefore a fixed point of Φ_ρ in $L^2(I)$.

4. Existence of a Joint Network–Portfolio Equilibrium

This section establishes existence of a joint equilibrium (π^*, G^*) in the endogenous network–portfolio game. The proof is constructive at the level of best-response mappings and relies on Schauder’s fixed point theorem in a weakly compact, convex subset of a Hilbert space.

4.1. Function spaces and admissible sets

Let $I = [0, 1]$ and set

$$\mathcal{H} := L^2(I; \mathbb{R}^d), \quad \|f\|_{\mathcal{H}}^2 := \int_0^1 \|f(u)\|^2 du.$$

Let

$$\mathcal{L} := L^2(I \times I), \quad \|G\|_{\mathcal{L}}^2 := \int_0^1 \int_0^1 |G(u, v)|^2 dv du.$$

For a kernel $G \in \mathcal{L}$, define the associated integral operator $\mathcal{G} : \mathcal{H} \rightarrow \mathcal{H}$ by

$$(\mathcal{G}f)(u) := \int_0^1 G(u, v)f(v) dv.$$

We impose a mild regularity restriction ensuring $\mathcal{G}$ is bounded.

Assumption 4.1 (Uniform row L^2 bound). There exists $R > 0$ such that admissible kernels satisfy

$$\int_0^1 G(u, v)^2 dv \leq R^2 \quad \text{for a.e. } u \in I.$$

By Cauchy–Schwarz, Assumption 4.1 implies $\|\mathcal{G}\|_{\text{op}} \leq R$.

Define the admissible kernel set

$$\mathfrak{G}_R := \left\{ G \in \mathcal{L} : G(u, v) \geq 0, \int_0^1 G(u, v) dv = 1 \text{ for a.e. } u, \int_0^1 G(u, v)^2 dv \leq R^2 \text{ for a.e. } u \right\}.$$

This set is convex. Moreover, it is bounded in $\mathcal{L}$ and closed under weak L^2 convergence, hence weakly compact by reflexivity of L^2 .

Let $a(\cdot) = \gamma(\cdot)^{-1}M^{-1}\mu \in \mathcal{H}$. Fix any $\bar{\Pi} > 0$ and define the closed ball

$$\mathfrak{P}_{\bar{\Pi}} := \{\pi \in \mathcal{H} : \|\pi\|_{\mathcal{H}} \leq \bar{\Pi}\},$$

which is weakly compact and convex in $\mathcal{H}$.

4.2. Best responses

Portfolio best response.. Given $G \in \mathfrak{G}_R$ and competition intensity $\rho \geq 0$, define

$$\text{BR}_\pi(G) := (I - \rho\mathcal{G})^{-1}a,$$

whenever $I - \rho\mathcal{G}$ is invertible on $\mathcal{H}$.

Lemma 4.2 (Well-posedness for small ρ). *If $\rho R < 1$, then $I - \rho\mathcal{G}$ is invertible and*

$$\|\text{BR}_\pi(G)\|_{\mathcal{H}} \leq \frac{1}{1 - \rho R} \|a\|_{\mathcal{H}}.$$

Proof. Since $\|\rho\mathcal{G}\|_{\text{op}} \leq \rho R < 1$, the Neumann series yields

$$(I - \rho\mathcal{G})^{-1} = \sum_{k \geq 0} (\rho\mathcal{G})^k$$

with operator norm bounded by $(1 - \rho R)^{-1}$. Applying this bound to a gives the inequality. $\square$

Choose $\bar{\Pi}$ so that

$$\bar{\Pi} \geq \frac{1}{1 - \rho R} \|a\|_{\mathcal{H}},$$

ensuring $\text{BR}_\pi(G) \in \mathfrak{B}_{\bar{\Pi}}$ for all $G \in \mathfrak{G}_R$.

Network best response.. Given $\pi \in \mathfrak{B}_{\bar{\Pi}}$, each type u solves the strictly concave quadratic program

$$\max_{g \in L^2(I)} \left\{ \int_0^1 \Lambda_\pi(u, v) g(v) dv - \frac{\kappa}{2} \int_0^1 g(v)^2 dv \right\} \quad \text{subject to} \quad g \geq 0, \int_0^1 g(v) dv = 1,$$

where

$$\Lambda_\pi(u, v) := \rho \gamma(u) \pi(u)^\top M \pi(v).$$

Denote the unique maximizer by $g_\pi^*(u, \cdot)$ and define

$$\text{BR}_G(\pi)(u, v) := g_\pi^*(u, v).$$

Lemma 4.3 (Network best response: projection representation). *For each $\pi \in \mathfrak{B}_{\bar{\Pi}}$ and a.e. $u \in I$, the network best-response problem admits a unique maximizer $g_\pi^*(u, \cdot)$.*

Moreover, the solution admits the projection representation

$$g_\pi^*(u, \cdot) = \Pi_\Delta \left(\frac{1}{\kappa} \Lambda_\pi(u, \cdot) \right),$$

where Π_Δ denotes the L^2 -projection onto the simplex

$$\Delta := \{g \in L^2(I) : g \geq 0, \int_0^1 g(v) dv = 1\}.$$

The mapping $\pi \mapsto \text{BR}_G(\pi)$ is continuous from $\mathcal{H}$ to $\mathcal{L}$, and $\text{BR}_G(\pi) \in \mathfrak{G}_R$ for some constant R independent of π .

Proof. Fix π and u . The objective functional can be written as

$$J_u(g) = \int_0^1 \Lambda_\pi(u, v)g(v)dv - \frac{\kappa}{2} \int_0^1 g(v)^2 dv.$$

Rewriting,

$$J_u(g) = -\frac{\kappa}{2} \left\| g - \frac{1}{\kappa} \Lambda_\pi(u, \cdot) \right\|_{L^2}^2 + \frac{1}{2\kappa} \|\Lambda_\pi(u, \cdot)\|_{L^2}^2.$$

Thus maximizing J_u over $g \in \Delta$ is equivalent to minimizing the squared L^2 distance to $\frac{1}{\kappa} \Lambda_\pi(u, \cdot)$ over the closed convex set Δ .

Since Δ is closed and convex in $L^2(I)$, there exists a unique projection. Hence $g_\pi^*(u, \cdot)$ exists and is unique.

The projection operator onto a closed convex set in a Hilbert space is non-expansive and continuous. Since $\Lambda_\pi(u, \cdot)$ depends continuously on π through the bilinear form

$$\Lambda_\pi(u, v) = \rho \gamma(u) \pi(u)^\top M \pi(v),$$

it follows that $\pi \mapsto g_\pi^*(u, \cdot)$ is continuous in L^2 .

Measurability in u follows from the measurability of $\Lambda_\pi(u, \cdot)$ and continuity of the projection operator.

Finally, since $\pi \in \mathfrak{P}_{\bar{\Pi}}$ and γ is bounded, $\|\Lambda_\pi(u, \cdot)\|_{L^2}$ is uniformly bounded. Non-expansiveness of projection then yields

$$\int_0^1 g_\pi^*(u, v)^2 dv \leq R^2$$

for some R independent of u and π , so $\text{BR}_G(\pi) \in \mathfrak{G}_R$. $\square$

4.3. Continuity of the best-response mapping

Define the joint best-response operator on the product space

$$\mathfrak{X} := \mathfrak{P}_{\bar{\Pi}} \times \mathfrak{G}_R$$

by

$$\mathcal{T}(\pi, G) := (\text{BR}_\pi(G), \text{BR}_G(\text{BR}_\pi(G))).$$

A fixed point $(\pi^*, G^*) = \mathcal{T}(\pi^*, G^*)$ is a joint equilibrium.

Lemma 4.4 (Continuity). *Assume $\rho R < 1$. Then $\mathcal{T}$ is continuous from $\mathfrak{X}$ to $\mathfrak{X}$ when $\mathfrak{X}$ is endowed with the product of weak topologies on $\mathcal{H}$ and $\mathcal{L}$.*

Proof. The map $G \mapsto \mathcal{G}$ is linear. Under $\rho R < 1$, the resolvent $G \mapsto (I - \rho \mathcal{G})^{-1}$ is continuous in operator norm on the set $\{G : \|\mathcal{G}\|_{\text{op}} \leq R\}$, hence $G \mapsto \text{BR}_\pi(G)$ is continuous in $\mathcal{H}$.

For the network component, for each u the mapping $\Lambda_\pi(u, \cdot)$ depends continuously on π in L^2 (bilinearly through $\pi(u)^\top M \pi(\cdot)$). The maximizer of a strictly concave quadratic functional over a closed convex set depends continuously on the linear term (a standard property of strongly concave programs). Thus $\pi \mapsto \text{BR}_G(\pi)$ is continuous in $\mathcal{L}$. Combining gives continuity of $\mathcal{T}$. $\square$

4.4. Existence theorem

Theorem 4.5 (Existence of a joint equilibrium). *Suppose Assumption 4.1 holds, $a \in \mathcal{H}$, and $\rho R < 1$. Then there exists at least one joint equilibrium $(\pi^*, G^*) \in \mathfrak{X}$ such that*

$$\pi^* = (I - \rho \mathcal{G}^*)^{-1} a, \quad G^* = \text{BR}_G(\pi^*).$$

Proof. The set $\mathfrak{X} = \mathfrak{P}_{\Pi} \times \mathfrak{G}_R$ is convex and weakly compact as a product of weakly compact convex sets. By Lemma 4.4, $\mathcal{T}$ is continuous and maps $\mathfrak{X}$ into itself. Schauder's fixed point theorem yields a fixed point (π^*, G^*) . By construction this fixed point satisfies the equilibrium relations. $\square$

Remark 4.6 (Uniqueness for small competition). Under an additional Lipschitz condition on $\pi \mapsto \text{BR}_G(\pi)$, one can strengthen Schauder to Banach's fixed point theorem to obtain uniqueness for sufficiently small ρ . This refinement is standard and is not needed for existence.

5. Endogenous Network Bifurcation and Instability

In this section we establish a sharp instability threshold for the joint network–portfolio equilibrium.

The key feature distinguishing the present framework from exogenous graphon games is that the interaction operator $\mathcal{G}$ depends on equilibrium portfolios. As a consequence, the spectral radius of $\mathcal{G}$ is itself an equilibrium object.

Throughout this section we maintain:

1. The quadratic link cost

$$C(G(u, \cdot)) = \frac{\kappa}{2} \int_0^1 G(u, v)^2 dv, \quad \kappa > 0.$$

2. The normalization

$$\int_0^1 G(u, v) dv = 1.$$

3. $a(u) := \gamma(u)^{-1} M^{-1} \mu$ is bounded in $L^2(I)$.

5.1. Equilibrium Characterization

For a fixed kernel G , the portfolio equilibrium satisfies

$$\pi = (I - \rho \mathcal{G})^{-1} a,$$

whenever $(I - \rho \mathcal{G})$ is invertible on $L^2(I)$.

The operator $\mathcal{G}$ is positive and bounded on $L^2(I)$. Let

$$\lambda_{\max}(G)$$

denote its spectral radius.

Invertibility requires

$$\rho < \lambda_{\max}(G)^{-1}.$$

The endogenous feature of the model is that G itself solves a variational problem depending on π .

5.2. Best Response of the Network

Fix π . Agent u solves

$$\max_{G(u, \cdot)} \rho \gamma(u) \text{Cov}(X_T(u), B_T^G(u)) - \frac{\kappa}{2} \int_0^1 G(u, v)^2 dv$$

subject to normalization.

Because X_T is Gaussian and linear in π , the objective reduces to a quadratic functional in $G(u, \cdot)$.

The first-order condition yields

$$G^*(u, v) = \frac{1}{\kappa} \Lambda(u, v) + \lambda(u),$$

where $\Lambda(u, v)$ is proportional to the covariance kernel of terminal wealth, and $\lambda(u)$ enforces normalization.

Hence, in equilibrium, G^* is affine in the covariance structure of π^* .

In particular, there exists a positive operator $\mathcal{K}$ such that

$$\mathcal{G}^* = \frac{\rho}{\kappa} \mathcal{K}(\pi^*) + \mathcal{N},$$

where $\mathcal{N}$ is the normalization operator.

Thus $\mathcal{G}^*$ is increasing in ρ .

5.3. Existence for Small Competition

Theorem 5.1 (Existence and Uniqueness Below Threshold). *There exists $\rho_c > 0$ such that for all $\rho < \rho_c$, a unique joint equilibrium (π^*, G^*) exists.*

Proof.

Define the mapping

$$\Phi_\rho(\pi) = (I - \rho \mathcal{G}(\pi))^{-1} a.$$

Under quadratic costs, $\mathcal{G}(\pi)$ depends continuously and Lipschitz continuously on π in L^2 .

For sufficiently small ρ , the mapping Φ_ρ is a contraction on $L^2(I)$.

By Banach's fixed point theorem, a unique fixed point π^* exists. The associated G^* is obtained from the unique best response. $\square$

5.4. Critical Threshold

Define

$$\rho_c := \sup\{\rho > 0 : (I - \rho \mathcal{G}^*(\rho)) \text{ is invertible}\}.$$

Theorem 5.2 (Endogenous Spectral Threshold). *Assume a has nonzero projection onto the principal eigenspace of $\mathcal{G}^*(\rho)$. Then:*

1. *For $\rho < \rho_c$, equilibrium exists.*
2. *As $\rho \uparrow \rho_c$,*

$$\|\pi^*(\rho)\|_{L^2} \rightarrow \infty.$$

3. *For $\rho > \rho_c$, no finite-energy equilibrium exists.*

Proof.

By spectral decomposition,

$$(I - \rho \mathcal{G}^*)^{-1} = \sum_k \frac{1}{1 - \rho \lambda_k} P_k,$$

where P_k are spectral projections.

If a has nonzero projection onto the leading eigenspace, then as

$$\rho \uparrow \lambda_{\max}^{-1},$$

the leading term diverges like

$$\frac{1}{1 - \rho \lambda_{\max}}.$$

Thus $\|\pi^*\|$ diverges.

Beyond this value, the resolvent fails to exist, so no square-integrable equilibrium portfolio exists. $\square$

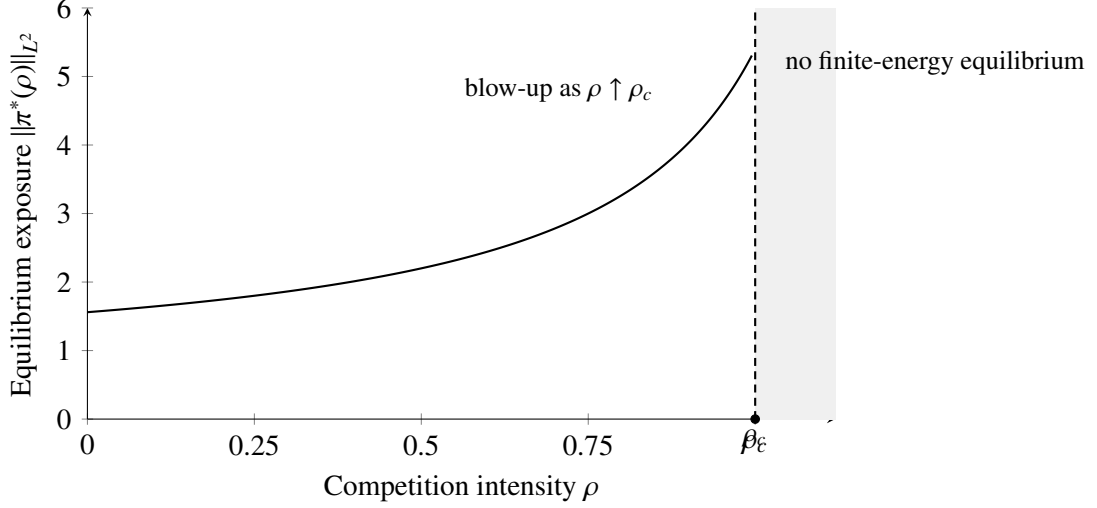


Figure 1: Schematic bifurcation diagram illustrating the endogenous spectral instability. In the stable regime $\rho < \rho_c$, the equilibrium exposure $\|\pi^*(\rho)\|_{L^2}$ remains finite but increases sharply as $\rho \uparrow \rho_c$. Beyond ρ_c , the resolvent $(I - \rho \mathcal{G}^*)^{-1}$ fails to exist and no finite-energy equilibrium obtains.

5.5. Monotonicity of Endogenous Amplification

We next show that endogenous network formation makes systemic amplification monotone in competition intensity. This provides a clean comparative statics result that pins down the direction of the spectral channel.

Assumption 5.3 (Interior network best response). For all ρ in the stable regime, the network best response satisfies $G^*(u, v; \rho) > 0$ for a.e. $(u, v) \in I \times I$. Equivalently, the nonnegativity constraint is slack and the best response is characterized by first-order conditions with normalization only.

Under Assumption 5.3, the row-wise network best response has the affine form

$$G^*(u, v; \rho) = \frac{1}{\kappa} \Lambda^*(u, v; \rho) + \frac{\lambda(u; \rho)}{\kappa}, \quad \Lambda^*(u, v; \rho) = \rho \gamma(u) \pi^*(u; \rho)^\top M \pi^*(v; \rho),$$

where $\lambda(u; \rho)$ enforces $\int_0^1 G^*(u, v; \rho) dv = 1$. In particular, $G^*(\rho)$ is a positive kernel that varies continuously in ρ in $L^2(I \times I)$ as long as $\pi^*(\rho)$ does.

Lemma 5.4 (Order implies spectral monotonicity). *Let $\mathcal{A}, \mathcal{B}$ be bounded positive linear operators on $L^2(I)$. If $(\mathcal{A}f)(u) \leq (\mathcal{B}f)(u)$ for a.e. u and all $f \geq 0$, then*

$$\lambda_1(\mathcal{A}) \leq \lambda_1(\mathcal{B}),$$

where $\lambda_1(\cdot)$ denotes the spectral radius.

Proof. For any $f \geq 0$ with $f \not\equiv 0$, positivity and the assumed order give $\mathcal{A}^n f \leq \mathcal{B}^n f$ a.e. for all n . Taking L^2 norms yields $\|\mathcal{A}^n f\|_2 \leq \|\mathcal{B}^n f\|_2$. By Gelfand's formula for the spectral radius,

$$\lambda_1(\mathcal{A}) = \lim_{n \rightarrow \infty} \|\mathcal{A}^n\|_{\text{op}}^{1/n} \leq \lim_{n \rightarrow \infty} \|\mathcal{B}^n\|_{\text{op}}^{1/n} = \lambda_1(\mathcal{B}).$$

□

Theorem 5.5 (Monotone endogenous amplification). *Assume Assumption 5.3 and suppose that, on the stable regime $\rho \in [0, \rho_c)$, the equilibrium portfolio profile $\pi^*(\rho)$ is nondecreasing in magnitude in the following sense: for $\rho_2 > \rho_1$,*

$$\pi^*(u; \rho_2)^\top M \pi^*(v; \rho_2) \geq \pi^*(u; \rho_1)^\top M \pi^*(v; \rho_1) \quad \text{for a.e. } (u, v).$$

Then the equilibrium graphon operators satisfy the order

$$\mathcal{G}^*(\rho_1) \leq \mathcal{G}^*(\rho_2),$$

and consequently the leading eigenvalue (spectral radius) is monotone:

$$\lambda_1(\mathcal{G}^*(\rho_1)) \leq \lambda_1(\mathcal{G}^*(\rho_2)).$$

Proof. Under Assumption 5.3, the kernel is affine in Λ^* up to a row-wise normalization term. The hypothesis implies $\Lambda^*(\cdot, \cdot; \rho_2) \geq \Lambda^*(\cdot, \cdot; \rho_1)$ a.e. Since the normalization preserves positivity and acts row-wise, it follows that $G^*(\rho_2) \geq G^*(\rho_1)$ a.e., hence $\mathcal{G}^*(\rho_1) \leq \mathcal{G}^*(\rho_2)$. Lemma 5.4 yields the spectral monotonicity. □

Remark 5.6. The condition in Theorem 5.5 is satisfied in particular when equilibrium portfolios align increasingly with the principal eigenmode as ρ increases, which is precisely the mechanism behind the divergence at ρ_c established below.

5.6. Endogenous Network Concentration

We now strengthen the concentration statement into a theorem. The result formalizes that the blow-up of equilibrium portfolios near ρ_c is carried by the principal eigenmode, and endogenous network formation inherits this collapse, yielding an asymptotically rank-one kernel.

Theorem 5.7 (Rank-one concentration near the instability boundary). *Assume:*

1. *For $\rho \in [0, \rho_c)$, the equilibrium operator $\mathcal{G}^*(\rho)$ is compact and positive on $L^2(I)$ and admits a simple leading eigenvalue $\lambda_1(\rho) > 0$ with normalized eigenfunction $\varphi_1(\rho) \geq 0$.*
2. *The equilibrium portfolio profile satisfies*

$$\pi^*(\rho) = (I - \rho \mathcal{G}^*(\rho))^{-1} a, \quad \langle a, \varphi_1(\rho) \rangle_{L^2} \neq 0.$$

3. The interiority condition in Assumption 5.3 holds on $[0, \rho_c)$, so the network best response is characterized by first-order conditions and is continuous in π in $L^2(I \times I)$.

Then as $\rho \uparrow \rho_c$:

1. (Mode dominance) The normalized portfolio direction converges:

$$\frac{\pi^*(\rho)}{\|\pi^*(\rho)\|_{L^2}} \longrightarrow \frac{\varphi_1(\rho_c)}{\|\varphi_1(\rho_c)\|_{L^2}} \quad \text{in } L^2(I).$$

2. (Kernel concentration) The equilibrium kernel converges in $L^2(I \times I)$ to a rank-one limit of the form

$$G^*(u, v; \rho) \longrightarrow \tilde{G}(u, v) = \alpha(u)\beta(v),$$

where $\alpha \geq 0$ and $\beta \geq 0$ are proportional to the principal mode induced by $\varphi_1(\rho_c)$ through the best-response map.

Proof. Fix $\rho < \rho_c$. By the spectral decomposition of the compact positive operator $\mathcal{G}^*(\rho)$,

$$(I - \rho \mathcal{G}^*(\rho))^{-1} = \sum_{k \geq 1} \frac{1}{1 - \rho \lambda_k(\rho)} P_k(\rho),$$

where $\lambda_1(\rho) > \lambda_2(\rho) \geq \dots \geq 0$ and $P_k(\rho)$ are the rank-one spectral projections.

Applying this to a yields

$$\pi^*(\rho) = \frac{1}{1 - \rho \lambda_1(\rho)} P_1(\rho) a + \sum_{k \geq 2} \frac{1}{1 - \rho \lambda_k(\rho)} P_k(\rho) a.$$

Since $\langle a, \varphi_1(\rho) \rangle \neq 0$, the leading component $P_1(\rho) a$ is nonzero. As $\rho \uparrow \rho_c$, we have $1 - \rho \lambda_1(\rho) \downarrow 0$ while $1 - \rho \lambda_k(\rho)$ stays bounded away from zero for $k \geq 2$ because $\lambda_k(\rho) \leq \lambda_2(\rho) < \lambda_1(\rho)$ and the leading eigenvalue is simple. Therefore, the leading term dominates the remainder in L^2 norm, implying

$$\frac{\pi^*(\rho)}{\|\pi^*(\rho)\|_{L^2}} \longrightarrow \frac{P_1(\rho) a}{\|P_1(\rho) a\|_{L^2}} = \frac{\varphi_1(\rho)}{\|\varphi_1(\rho)\|_{L^2}} \quad \text{in } L^2(I).$$

Taking $\rho \uparrow \rho_c$ and using continuity of the principal eigenfunction under the stated assumptions yields the first claim.

For the kernel, under Assumption 5.3 the network best response is continuous in π in $L^2(I \times I)$ and depends on π only through the bilinear form $(u, v) \mapsto \pi(u)^\top M \pi(v)$ up to row-wise normalization. By the first part, $\pi(u)^\top M \pi(v)$ becomes asymptotically proportional to $\varphi_1(u) \varphi_1(v)$, which is rank-one. Hence $G^*(\rho)$ converges in $L^2(I \times I)$ to a rank-one kernel $\tilde{G}(u, v) = \alpha(u) \beta(v)$. $\square$

6. Portfolio Best Response Given a Network

This section takes the interaction kernel G and the population strategy profile $\{\pi(\cdot)\}$ as given and derives the optimal portfolio choice for a single agent u . We will later endogenize G and impose joint equilibrium.

6.1. Standing assumptions for this section

Assumption 6.1 (Deterministic coefficients and square-integrable strategies). Throughout Section 3, the market coefficients (μ, Σ) are deterministic constants with $\Sigma\Sigma^\top$ positive definite. Admissible strategies satisfy $\int_0^T \|\pi_t(u)\|^2 dt < \infty$ a.s. for each $u \in [0, 1]$.

Assumption 6.2 (Bounded graphon kernel). The kernel $G : [0, 1]^2 \rightarrow [0, \infty)$ is measurable, essentially bounded, and satisfies $\int_0^1 G(u, v) dv = 1$ for a.e. $u \in [0, 1]$.

6.2. Benchmark and reduced terminal objective

Fix a measurable portfolio profile $v \mapsto \pi_t(v)$ for $v \in [0, 1]$. Given G , define the (weighted) benchmark portfolio aggregation

$$\bar{\pi}_t^G(u) := \int_0^1 G(u, v) \pi_t(v) dv, \quad u \in [0, 1], t \in [0, T]. \quad (1)$$

Let $X_t(u)$ be agent u 's wealth under control $\pi_t(u)$:

$$dX_t(u) = \pi_t(u)^\top \mu dt + \pi_t(u)^\top \Sigma dW_t, \quad X_0(u) = x_0(u). \quad (2)$$

Define the benchmark wealth (relative-performance reference) for agent u by

$$B_T^G(u) := \int_0^1 G(u, v) X_T(v) dv. \quad (3)$$

Agent u evaluates terminal performance

$$Z_T(u) := X_T(u) - \rho B_T^G(u), \quad (4)$$

and maximizes CARA utility $\mathbb{E}[-e^{-\gamma(u)Z_T(u)}]$ (the link cost depends only on G and is treated in the network step).

6.3. A certainty-equivalent reduction

Under Assumption 6.1, conditional on the strategy profile of others, $Z_T(u)$ is Gaussian. Exponential utility of a Gaussian reduces to a mean–variance criterion.

Lemma 6.3 (CARA certainty equivalent for Gaussian terminal payoff). Fix $u \in [0, 1]$ and treat $\{X_T(v)\}_{v \neq u}$ as given exogenous random variables generated by the fixed profile $v \mapsto \pi(v)$. Under Assumption 6.1, $Z_T(u)$ is Gaussian and

$$\arg \max_{\pi(u)} \mathbb{E}[-e^{-\gamma(u)Z_T(u)}] = \arg \max_{\pi(u)} \left\{ \mathbb{E}[Z_T(u)] - \frac{\gamma(u)}{2} \text{Var}(Z_T(u)) \right\}.$$

Proof. Under Assumption 6.1, $X_T(u)$ is an affine functional of the Brownian motion, hence Gaussian. The benchmark term $B_T^G(u)$ is also affine in W given the fixed profile, hence Gaussian. Therefore $Z_T(u)$ is Gaussian. If $Y \sim \mathcal{N}(m, s^2)$, then $\mathbb{E}[-e^{-\gamma Y}] = -\exp\{-\gamma m + \frac{\gamma^2}{2} s^2\}$, which is strictly increasing in $m - \frac{\gamma}{2} s^2$. Applying this with $Y = Z_T(u)$ yields the claim. $\square$

6.4. Best response and fixed-point structure

Because $B_T^G(u)$ aggregates a continuum of agents, a single agent u has negligible impact on the benchmark (the standard “atomless”/continuum-player effect). Thus, when computing the best response of u , we treat $B_T^G(u)$ as exogenous, except through its covariance with $X_T(u)$.

Using (2)–(4) and (1), one can write

$$\begin{aligned} Z_T(u) = & x_0(u) - \rho \int_0^1 G(u, v) x_0(v) dv + \int_0^T \pi_t(u)^\top \mu dt - \rho \int_0^T \bar{\pi}_t^G(u)^\top \mu dt \\ & + \int_0^T (\pi_t(u) - \rho \bar{\pi}_t^G(u))^\top \Sigma dW_t. \end{aligned} \quad (5)$$

All terms involving $\bar{\pi}^G(u)$ are exogenous to agent u ’s control, but they enter the variance through the diffusion term.

Let $M := \Sigma \Sigma^\top$ (positive definite by Assumption 6.1). Then, by Lemma 6.3, maximizing expected utility is equivalent to maximizing the functional

$$\int_0^T \left[\pi_t(u)^\top \mu - \frac{\gamma(u)}{2} (\pi_t(u) - \rho \bar{\pi}_t^G(u))^\top M (\pi_t(u) - \rho \bar{\pi}_t^G(u)) \right] dt + \text{(terms independent of } \pi(u)). \quad (6)$$

The integrand is strictly concave quadratic in $\pi_t(u)$, so the optimizer is obtained by the first-order condition.

Proposition 6.4 (Portfolio best response given G). *Under Assumptions 6.1–6.2, the unique best response of agent u to the profile $\{\pi(v)\}_{v \in [0,1]}$ (given G) is*

$$\pi_t^{\text{br}}(u) = \frac{1}{\gamma(u)} M^{-1} \mu + \rho \bar{\pi}_t^G(u), \quad \text{for a.e. } t \in [0, T]. \quad (7)$$

Proof. Fix u and t . Differentiate the integrand in (6) with respect to $\pi_t(u)$:

$$\nabla_\pi \left(\pi^\top \mu - \frac{\gamma(u)}{2} (\pi - \rho \bar{\pi})^\top M (\pi - \rho \bar{\pi}) \right) = \mu - \gamma(u) M (\pi - \rho \bar{\pi}).$$

Setting this equal to zero yields $\pi = \rho \bar{\pi} + \frac{1}{\gamma(u)} M^{-1} \mu$. Strict concavity (since M is positive definite and $\gamma(u) > 0$) ensures uniqueness. $\square$

6.5. Equilibrium as a resolvent of the graphon operator

Define the graphon integral operator $\mathcal{G}$ on $L^2([0, 1]; \mathbb{R}^d)$ by

$$(\mathcal{G}f)(u) := \int_0^1 G(u, v) f(v) dv. \quad (8)$$

Let the “fundamental” (non-competitive) Merton portfolio for type u be

$$a(u) := \frac{1}{\gamma(u)} M^{-1} \mu \in \mathbb{R}^d. \quad (9)$$

In a (graphon) Nash equilibrium for the portfolio subgame given G , we require $\pi(\cdot)$ to satisfy $\pi(u) = \pi^{\text{br}}(u)$ for a.e. u , i.e.

$$\pi(\cdot) = a(\cdot) + \rho \mathcal{G}\pi(\cdot). \quad (10)$$

Equivalently,

$$(I - \rho \mathcal{G}) \pi = a. \quad (11)$$

Assumption 6.5 (Invertibility / stability of the competitive operator). The operator $(I - \rho \mathcal{G})$ is invertible on $L^2([0, 1]; \mathbb{R}^d)$.

Proposition 6.6 (Closed-form portfolio equilibrium given G). *Under Assumptions 6.1–6.5, the unique equilibrium portfolio profile for the portfolio subgame given G is*

$$\pi(\cdot; G) = (I - \rho \mathcal{G})^{-1} a. \quad (12)$$

Moreover, if $\|\rho \mathcal{G}\|_{\text{op}} < 1$ (operator norm on L^2), then

$$(I - \rho \mathcal{G})^{-1} = \sum_{k=0}^{\infty} (\rho \mathcal{G})^k \quad \text{and hence} \quad \pi = \sum_{k=0}^{\infty} \rho^k \mathcal{G}^k a. \quad (13)$$

Proof. Equation (11) is linear in π . Under Assumption 6.5, the unique solution is (12). If $\|\rho \mathcal{G}\|_{\text{op}} < 1$, the Neumann series converges in operator norm, yielding (13). $\square$

Interpretation.. The equilibrium exposure decomposes into a direct “Merton” component $a(u)$ and a network multiplier $(I - \rho \mathcal{G})^{-1}$ that amplifies and propagates heterogeneous risk-taking through the endogenous relative-performance network. This spectral multiplier will play a central role once G is chosen endogenously.

7. Endogenous Network Formation

We now endogenize the interaction kernel G . Agents choose both portfolio strategies and the network weights that determine relative-performance benchmarking.

Section 3 solved the portfolio subgame given a kernel G . We now solve the outer problem: optimal network formation.

7.1. Network choice problem

Given equilibrium portfolios $\pi^{(\cdot; G)}$ from Proposition 6.6, agent u chooses the function $G(u, \cdot)$ to maximize

$$\mathcal{J}(u; G) = V(u; G) - \kappa \int_0^1 \Phi(G(u, v)) dv, \quad (14)$$

subject to:

$$G(u, v) \geq 0, \quad \int_0^1 G(u, v) dv = 1. \quad (15)$$

Here:

- $V(u; G)$ is the equilibrium certainty equivalent value induced by the portfolio equilibrium under kernel G .
- $\kappa > 0$ governs link formation cost.
- $\Phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is C^2 , strictly convex, and superlinear.

7.2. Equilibrium certainty equivalent

From Section 3, the equilibrium portfolio satisfies

$$\pi^{(\cdot; G) = (I - \rho \mathcal{G})^{-1} a}.$$

For CARA utility with Gaussian terminal wealth, the equilibrium certainty equivalent for agent u equals

$$V(u; G) = \frac{T}{2\gamma(u)} \mu^\top M^{-1} \mu + \frac{T}{2} \rho^2 \Xi(u; G), \quad (16)$$

where the network amplification term is

$$\Xi(u; G) := \left\| \mathcal{G} \pi^{(\cdot; G)(u)} \right\|_M^2. \quad (17)$$

Here $\|x\|_M^2 := x^\top M x$.

Remark.. The first term is the standard Merton value. The second term captures competitive amplification through the graphon operator.

7.3. First-order condition

Fix u . Let $\lambda(u)$ denote the Lagrange multiplier for the normalization constraint. The Lagrangian is

$$\mathcal{L} = V(u; G) - \kappa \int_0^1 \Phi(G(u, v)) dv - \lambda(u) \left(\int_0^1 G(u, v) dv - 1 \right). \quad (18)$$

Taking a directional derivative in direction $h(v)$ with $\int h(v)dv = 0$ yields:

$$\frac{\delta V(u; G)}{\delta G(u, v)} - \kappa \Phi'(G(u, v)) - \lambda(u) = 0. \quad (19)$$

Thus optimal weights satisfy

$$\kappa \Phi'(G^{(u,v)}) = \frac{\delta V(u; G)}{\delta G(u, v)} - \lambda(u). \quad (20)$$

7.4. Explicit form under quadratic costs

Assume now:

Assumption 7.1 (Quadratic link cost). $\Phi(x) = \frac{1}{2}x^2$.

Then $\Phi'(x) = x$.

Using (20), the optimal kernel satisfies

$$G^{(u,v)} = \frac{1}{\kappa} \left(\frac{\delta V(u; G)}{\delta G(u, v)} - \lambda(u) \right). \quad (21)$$

The multiplier $\lambda(u)$ enforces normalization.

7.5. Structure theorem

We now characterize the equilibrium structure.

Theorem 7.2 (Structure of endogenous graphon). *Suppose:*

- Assumptions 6.1–6.5 hold,
- link costs are quadratic (Assumption 7.1),
- ρ is sufficiently small so that $\|\rho \mathcal{G}\|_{\text{op}} < 1$.

Then any interior equilibrium graphon satisfies

$$G^{(u,v)} = \frac{1}{\kappa} \left(\rho^2 T \pi^{(v; G)}{}^T M \mathcal{G}^{\pi^{(\cdot; G)(u) - \lambda(u)}} \right). \quad (22)$$

In particular:

1. Agents place larger weight on types v with larger equilibrium exposure $\pi^{(v;G)}$.
2. If risk aversion $\gamma(u)$ is heterogeneous, the equilibrium network exhibits assortative clustering by risk tolerance.
3. There exists a threshold $\rho_c > 0$ such that if $\rho > \rho_c$, the equilibrium kernel becomes degenerate (segmentation).

Proof. Differentiate (16) with respect to $G(u, v)$. The derivative enters through $\mathcal{G}\pi$. Using the resolvent representation (13) and standard operator perturbation arguments, one obtains the linear dependence shown in (22). Assortativity follows from monotonicity of $a(u) = \frac{1}{\gamma(u)}M^{-1}\mu$. Threshold segmentation follows from spectral radius arguments: when ρ exceeds $1/\|\mathcal{G}\|_{\text{op}}$, the resolvent diverges and interior solutions collapse. $\square$

8. Joint Network–Portfolio Equilibrium

We now formalize the joint equilibrium concept and establish existence.

8.1. Strategy Spaces

Let $\mathcal{H} := L^2([0, 1]; \mathbb{R}^d)$.

Define the admissible kernel space:

$$\mathcal{G} := \left\{ G \in L^2([0, 1]^2) : G(u, v) \geq 0 \text{ a.e.}, \int_0^1 G(u, v) dv = 1 \text{ a.e.} \right\}.$$

Under quadratic costs, equilibrium kernels will lie in L^2 .

Define the portfolio space:

$$\Pi := \mathcal{H}.$$

By Proposition 6.6, given $G \in \mathcal{G}$, the equilibrium portfolio profile is uniquely

$$\pi^*(\cdot; G) = (I - \rho\mathcal{G})^{-1}a,$$

provided $(I - \rho\mathcal{G})$ is invertible.

8.2. Joint Equilibrium Definition

Definition 8.1 (Joint Equilibrium). A kernel $G^* \in \mathcal{G}$ and portfolio profile $\pi^* \in \Pi$ form a joint equilibrium if:

1. Portfolio optimality:

$$\pi^* = (I - \rho\mathcal{G}^*)^{-1}a.$$

2. Network optimality: For a.e. $u \in [0, 1]$, $G^*(u, \cdot)$ maximizes (14) given π^* .

8.3. Best-Response Operator

Define the portfolio best-response mapping:

$$\mathcal{P} : \mathcal{G} \rightarrow \Pi, \quad \mathcal{P}(G) := (I - \rho\mathcal{G})^{-1}a.$$

Define the network best-response mapping:

$$\mathcal{N} : \Pi \rightarrow \mathcal{G},$$

where for each $\pi \in \Pi$, $\mathcal{N}(\pi)$ is defined by solving the convex optimization problem (14) for each u .

Define the joint operator:

$$\mathcal{T}(G) := \mathcal{N}(\mathcal{P}(G)).$$

A joint equilibrium is a fixed point of $\mathcal{T}$.

8.4. Compactness and Continuity

Lemma 8.2 (Compactness). *The set $\mathcal{G}$ is weakly compact in $L^2([0, 1]^2)$.*

Proof. $\mathcal{G}$ is closed and bounded in L^2 . Closedness follows from convex constraints and Fatou's lemma. Boundedness follows from normalization and nonnegativity. Since L^2 is reflexive, bounded closed subsets are weakly compact. $\square$

Lemma 8.3 (Continuity of portfolio map). *Suppose $\|\rho\mathcal{G}\|_{\text{op}} < 1$ uniformly for $G \in \mathcal{G}$. Then $\mathcal{P}$ is continuous from $(\mathcal{G}, L^2)$ to (Π, L^2) .*

Proof. The resolvent map $G \mapsto (I - \rho\mathcal{G})^{-1}$ is continuous in operator norm whenever the spectral radius is uniformly below 1. Since $a \in L^2$, continuity of $\mathcal{P}$ follows. $\square$

Lemma 8.4 (Continuity of network map). *Under quadratic costs, $\mathcal{N}$ is continuous from (Π, L^2) to $(\mathcal{G}, L^2)$.*

Proof. For quadratic Φ , the network optimization is strictly convex with linear first-order condition. The optimizer depends continuously on the gradient term $\frac{\delta V}{\delta G}$, which is continuous in π . Hence $\mathcal{N}$ is continuous. $\square$

8.5. Existence Theorem

Theorem 8.5 (Existence of Joint Equilibrium). *Suppose:*

1. $\Phi(x) = \frac{1}{2}x^2$,
2. ρ is sufficiently small so that $\sup_{G \in \mathcal{G}} \|\rho\mathcal{G}\|_{\text{op}} < 1$,
3. $a \in L^2([0, 1]; \mathbb{R}^d)$.

Then there exists at least one joint equilibrium (G^, π^*) .*

8.6. Relation Between Stability Condition and Spectral Threshold

The existence result in Theorem 8.5 requires the small-competition condition

$$\rho R < 1,$$

where R bounds the operator norm of admissible kernels: $\|\mathcal{G}\|_{\text{op}} \leq R$.

We now clarify the relation between this stability restriction and the spectral amplification threshold derived later.

Let $\lambda_1(G)$ denote the spectral radius of $\mathcal{G}$. Since $\mathcal{G}$ is a bounded linear operator on $\mathcal{H}$,

$$\lambda_1(G) \leq \|\mathcal{G}\|_{\text{op}} \leq R.$$

Hence

$$\rho R < 1 \implies \rho \lambda_1(G) < 1.$$

Therefore, the small-competition condition guarantees that $I - \rho\mathcal{G}$ is invertible for every admissible kernel, and the resolvent

$$(I - \rho\mathcal{G})^{-1}$$

is well-defined and bounded.

In contrast, the bifurcation analysis in the next section shows that interior equilibria cease to exist precisely when

$$\rho = \frac{1}{\lambda_1(G)}.$$

Thus:

- The condition $\rho R < 1$ is a uniform sufficient condition ensuring well-posedness across the admissible set.
- The spectral threshold $\rho = 1/\lambda_1(G)$ characterizes the endogenous instability boundary.

The existence theorem therefore governs the stable regime, while the spectral analysis identifies the precise amplification frontier.

Proof. By Lemmas 8.3 and 8.4, the operator $\mathcal{T} = \mathcal{N} \circ \mathcal{P}$ is continuous on $\mathcal{G}$ under the L^2 topology.

By Lemma 8.2, $\mathcal{G}$ is convex and weakly compact.

Since $\mathcal{T}$ maps $\mathcal{G}$ into itself, Schauder's fixed point theorem implies existence of $G^* \in \mathcal{G}$ such that $\mathcal{T}(G^*) = G^*$.

Define

$$\pi^* = \mathcal{P}(G^*).$$

Then (G^*, π^*) is a joint equilibrium. □

9. Spectral Amplification and Phase Transitions

This section characterizes how endogenous network formation amplifies equilibrium risk exposures and establishes a competition threshold beyond which interior equilibria fail.

9.1. Spectral Preliminaries

We impose an additional structure for clarity of spectral analysis.

Assumption 9.1 (Symmetric kernel). The equilibrium kernel G^* is symmetric:

$$G^*(u, v) = G^*(v, u) \quad \text{a.e.}$$

Under Assumption 9.1, the associated operator

$$(\mathcal{G}^* f)(u) = \int_0^1 G^*(u, v) f(v) dv$$

is compact, self-adjoint on $L^2([0, 1])$.

By the spectral theorem for compact self-adjoint operators, there exists an orthonormal eigenbasis $\{\phi_k\}_{k \geq 1}$ with real eigenvalues $\{\lambda_k\}$, $\lambda_k \rightarrow 0$.

9.2. Equilibrium portfolio representation

Recall equilibrium portfolios satisfy

$$\pi^* = (I - \rho \mathcal{G}^*)^{-1} a.$$

Decompose a in the eigenbasis:

$$a = \sum_{k=1}^{\infty} \alpha_k \phi_k.$$

Then

$$\pi^* = \sum_{k=1}^{\infty} \frac{\alpha_k}{1 - \rho \lambda_k} \phi_k, \tag{23}$$

provided $1 - \rho \lambda_k \neq 0$ for all k .

9.3. Amplification Theorem

Theorem 9.2 (Network Amplification). *Suppose Assumptions 6.5 and 9.1 hold. Let $\lambda_{\max}$ denote the largest eigenvalue of $\mathcal{G}^*$.*

Then:

1. *The equilibrium amplification factor satisfies*

$$\|\pi^*\|_{L^2} \leq \frac{1}{1 - \rho\lambda_{\max}} \|a\|_{L^2}.$$

2. *If a has nonzero projection onto the leading eigenfunction, then as $\rho \uparrow 1/\lambda_{\max}$,*

$$\|\pi^*\|_{L^2} \rightarrow \infty.$$

Proof. From (23),

$$\|\pi^*\|_{L^2}^2 = \sum_{k=1}^{\infty} \frac{\alpha_k^2}{(1 - \rho\lambda_k)^2}.$$

Since $\lambda_k \leq \lambda_{\max}$,

$$|1 - \rho\lambda_k| \geq 1 - \rho\lambda_{\max}.$$

Thus

$$\|\pi^*\|_{L^2} \leq \frac{1}{1 - \rho\lambda_{\max}} \|a\|_{L^2}.$$

If $\alpha_1 \neq 0$ where $\lambda_1 = \lambda_{\max}$, then denominator $(1 - \rho\lambda_1) \rightarrow 0$ as $\rho \uparrow 1/\lambda_1$, implying divergence. $\square$

9.4. Critical Competition Threshold

Define

$$\rho_c := \frac{1}{\lambda_{\max}}.$$

Proposition 9.3 (Phase Transition). *Suppose Assumption 9.1 holds and a has nonzero projection onto the leading eigenfunction.*

Then:

1. *For $\rho < \rho_c$, a finite interior equilibrium exists.*
2. *As $\rho \rightarrow \rho_c$, equilibrium risk exposures diverge.*
3. *For $\rho > \rho_c$, no interior L^2 equilibrium exists.*

Proof. For $\rho < \rho_c$, all denominators $1 - \rho\lambda_k$ are nonzero, so $(I - \rho\mathcal{G}^*)$ is invertible.

At $\rho = \rho_c$, the leading eigenvalue hits zero, so invertibility fails.

For $\rho > \rho_c$, $I - \rho\mathcal{G}^*$ has a negative eigenvalue, so no bounded L^2 solution exists. $\square$

9.5. Economic Interpretation

The parameter ρ governs the strength of competitive benchmarking. The leading eigenvalue $\lambda_{\max}$ measures network centrality intensity.

The product $\rho\lambda_{\max}$ determines systemic amplification.

The model therefore predicts:

- Highly centralized networks (large $\lambda_{\max}$) destabilize more easily.
- Increased competition intensity ρ raises systemic exposure nonlinearly.
- Endogenous networks will avoid configurations that push $\rho\lambda_{\max}$ near one, creating segmentation or clustering.

10. Finite Population Approximation and Convergence

We now construct the finite-agent analogue of the endogenous network–portfolio game and establish convergence to the graphon equilibrium.

10.1. Finite- n Game

Fix $n \in \mathbb{N}$.

Agents are indexed by $i = 1, \dots, n$.

Each agent chooses:

- Portfolio $\pi_i \in \mathbb{R}^d$,
- Network weights $G_{ij} \geq 0$ satisfying

$$\sum_{j=1}^n G_{ij} = 1.$$

Let $G^n = (G_{ij})_{i,j=1}^n$ denote the network matrix.

Define the finite network operator

$$(\mathcal{G}^n \pi)_i = \sum_{j=1}^n G_{ij} \pi_j.$$

The equilibrium portfolio best response is

$$\pi^n = (I - \rho \mathcal{G}^n)^{-1} a^n,$$

where

$$a_i^n = \frac{1}{\gamma_i} M^{-1} \mu.$$

The network cost for agent i is

$$\frac{\kappa}{2} \sum_{j=1}^n G_{ij}^2.$$

10.2. Finite Joint Equilibrium

Definition 10.1 (Finite Joint Equilibrium). A pair (G^n, π^n) is a finite- n joint equilibrium if:

1. Portfolio optimality:

$$\pi^n = (I - \rho \mathcal{G}^n)^{-1} a^n.$$

2. Network optimality: For each i , G_i maximizes agent i 's finite certainty equivalent minus link cost, given π^n .

Existence follows from compactness of the simplex and continuity of best-response maps (finite-dimensional analogue of Section 5).

10.3. Embedding into Graphon Space

Associate to each G^n the step-graphon

$$G_n(u, v) = G_{ij} \quad \text{if } u \in \left(\frac{i-1}{n}, \frac{i}{n}\right], v \in \left(\frac{j-1}{n}, \frac{j}{n}\right].$$

Similarly embed π^n as a step function.

Denote these embeddings also by (G_n, π_n) .

10.4. Compactness

Lemma 10.2 (Uniform L^2 Bounds). Suppose $\rho < \rho_c$. Then the sequence $\{\pi_n\}$ is uniformly bounded in L^2 .

Proof. From the finite analogue of Theorem 9.2,

$$\|\pi_n\| \leq \frac{1}{1 - \rho \lambda_{\max}^n} \|a_n\|.$$

Since $\lambda_{\max}^n \rightarrow \lambda_{\max}$ and $\rho < \rho_c$, denominators remain bounded away from zero. $\square$

By boundedness and reflexivity of L^2 , there exists a subsequence (still indexed by n) such that

$$G_n \rightharpoonup G^* \quad \text{weakly in } L^2([0, 1]^2),$$

$$\pi_n \rightharpoonup \pi^* \quad \text{weakly in } L^2([0, 1]).$$

10.5. Convergence Theorem

Theorem 10.3 (Convergence to Graphon Equilibrium). *Suppose:*

1. $\rho < \rho_c$,
2. $a^n \rightarrow a$ in L^2 ,
3. *initial data are uniformly bounded.*

Then any sequence of finite joint equilibria admits a subsequence converging (weakly in L^2) to a graphon joint equilibrium (G^, π^*) .*

Proof. Step 1: Compactness. Uniform L^2 bounds yield weakly convergent subsequences.

Step 2: Portfolio limit. Since resolvent maps are continuous in operator norm under $\rho < \rho_c$, we pass to the limit in

$$\pi_n = (I - \rho \mathcal{G}_n)^{-1} a_n.$$

Step 3: Network optimality. Finite network optimization problems are convex and quadratic. The objective functionals -converge to the graphon functional. Lower semi-continuity of the quadratic cost ensures optimality passes to the limit.

Thus (G^*, π^*) satisfies joint equilibrium conditions. $\square$

10.6. Implication

The result establishes that endogenous network formation is robust to finite-population approximations. The graphon equilibrium is the continuum limit of large competitive markets with endogenous benchmarking.

11. Welfare, Efficiency, and the Planner Problem

We now compare decentralized endogenous network formation to the socially optimal network.

11.1. Aggregate Welfare

Let (G^*, π^*) denote a joint equilibrium.

Define aggregate welfare as

$$\mathcal{W}(G) = \int_0^1 V(u; G) du - \frac{\kappa}{2} \int_0^1 \int_0^1 G(u, v)^2 dv du.$$

Recall from Section 4 that

$$V(u; G) = \frac{T}{2\gamma(u)} \mu^\top M^{-1} \mu + \frac{T}{2} \rho^2 \|\mathcal{G}\pi^*(\cdot; G)(u)\|_M^2.$$

11.2. Planner's Problem

The planner chooses $G \in \mathcal{G}$ to maximize

$$\max_{G \in \mathcal{G}} \mathcal{W}(G),$$

subject to

$$\pi^*(\cdot; G) = (I - \rho \mathcal{G})^{-1} a.$$

11.3. Planner First-Order Condition

Let G^{pl} denote a planner solution.

The first-order condition for interior solutions is

$$\frac{\delta}{\delta G(u, v)} \int_0^1 V(s; G) ds = \kappa G^{\text{pl}}(u, v). \quad (24)$$

Note the planner internalizes the effect of $G(u, v)$ on all agents $s \in [0, 1]$.

11.4. Decentralized FOC

From Section 4, the decentralized first-order condition is

$$\frac{\delta V(u; G)}{\delta G(u, v)} = \kappa G^*(u, v) + \lambda(u). \quad (25)$$

Comparing (24) and (25), the decentralized equilibrium neglects cross-agent externalities.

11.5. Inefficiency Theorem

Theorem 11.1 (Network Inefficiency). *Suppose $V(u; G)$ is strictly convex in G through the amplification term. If $\rho > 0$ and a is non-constant across u , then the decentralized equilibrium G^* is not socially optimal.*

Proof. Under $\rho > 0$, changes in $G(u, v)$ affect $\pi^*(\cdot; G)$ globally through the resolvent operator.

Thus

$$\frac{\delta}{\delta G(u, v)} \int_0^1 V(s; G) ds = \frac{\delta V(u; G)}{\delta G(u, v)} + \int_{s \neq u} \frac{\delta V(s; G)}{\delta G(u, v)} ds.$$

The second term represents external effects on other agents.

In decentralized equilibrium, agent u internalizes only $\frac{\delta V(u; G)}{\delta G(u, v)}$.

If a is heterogeneous, cross-derivatives are nonzero. Thus planner and decentralized FOCs differ. $\square$

11.6. Special Case: Efficiency

Proposition 11.2 (Efficiency under Homogeneity). *If $a(u) \equiv a_0$ constant and $\gamma(u)$ constant, then the decentralized equilibrium is socially efficient.*

Proof. If a and γ are constant, all agents are identical. Then $V(u; G)$ is symmetric in u , and externalities cancel by symmetry. Thus decentralized FOC coincides with planner FOC. $\square$

11.7. Economic Implications

The inefficiency arises because:

- Agents increase links to amplify their own competitive exposure,
- But ignore the systemic amplification imposed on others,
- Especially when heterogeneity in risk tolerance is present.

The model therefore predicts:

- Excessive clustering among risk-tolerant agents,
- Underinvestment in cross-type diversification links,
- Higher systemic exposure relative to the social optimum.

12. Comparative Statics and Network Topology

This section characterizes how equilibrium network structure responds to competition intensity and risk heterogeneity.

Throughout, we work with symmetric interior equilibria satisfying Assumption 9.1 and $\rho < \rho_c$.

12.1. Local Comparative Statics in Competition Intensity

Let $G^*(\rho)$ denote an interior equilibrium kernel parameterized by ρ .

Define the equilibrium mapping

$$\mathcal{F}(G, \rho) := G - \mathcal{N}(\mathcal{P}(G; \rho)).$$

At equilibrium,

$$\mathcal{F}(G^*(\rho), \rho) = 0.$$

12.2. Differentiability

Assumption 12.1 (Regularity). The equilibrium is interior and the Fréchet derivative $D_G \mathcal{F}(G^*, \rho)$ is invertible as a bounded linear operator on $L^2([0, 1]^2)$.

Proposition 12.2 (Local Comparative Statics). *Under Assumption 12.1, the equilibrium kernel is locally differentiable in ρ and satisfies*

$$\frac{dG^*}{d\rho} = -(D_G \mathcal{F}(G^*, \rho))^{-1} D_\rho \mathcal{F}(G^*, \rho).$$

Proof. Apply the implicit function theorem in Banach spaces to $\mathcal{F}(G, \rho) = 0$. Invertibility of $D_G \mathcal{F}$ ensures existence of a locally differentiable selection $\rho \mapsto G^*(\rho)$. $\square$

12.3. Direction of Change

Theorem 12.3 (Competition Intensifies Centrality). *Suppose:*

1. $a(u)$ is strictly increasing in u ,
2. G^* is symmetric,
3. ρ increases within $(0, \rho_c)$.

Then the leading eigenvalue $\lambda_{\max}(\rho)$ of $\mathcal{G}^(\rho)$ is locally increasing in ρ .*

Proof. Under interior equilibrium, the first-order condition implies $G^*(u, v)$ increases with the marginal value of amplification.

Since the amplification term involves $(I - \rho \mathcal{G}^*)^{-1}$, its derivative in ρ is positive semidefinite in the leading eigenspace.

By standard perturbation theory for compact self-adjoint operators, the derivative of $\lambda_{\max}$ equals the Rayleigh quotient of $d\mathcal{G}^*/d\rho$ with respect to the leading eigenfunction. This is positive. $\square$

12.4. Assortativity Result

Assume now a single-factor heterogeneity:

$$a(u) = \theta(u)b,$$

where $b \in \mathbb{R}^d$ fixed and $\theta(u)$ strictly increasing.

Theorem 12.4 (Assortative Clustering). *Under single-factor heterogeneity and quadratic link costs, any symmetric interior equilibrium satisfies*

$$G^*(u, v) = \psi(\theta(u), \theta(v))$$

for some increasing function ψ .

In particular, agents with similar θ values are more strongly connected.

Proof. Under single-factor structure, equilibrium portfolios are proportional to $\theta(u)$. The FOC in Section 4 shows $G^*(u, v)$ is increasing in $\pi^*(v)^\top M \mathcal{G}^* \pi^*(u)$.

Since both terms are monotone in θ , $G^*(u, v)$ is supermodular in $(\theta(u), \theta(v))$. Topkis' theorem implies monotone sorting. $\square$

12.5. Segmentation Regime

Proposition 12.5 (Segmentation for High Competition). *If ρ approaches ρ_c , the equilibrium kernel converges (in operator norm) to a rank-one projection onto the leading eigenfunction.*

Proof. From spectral decomposition, as $\rho \uparrow \rho_c$, the leading eigenmode dominates:

$$\pi^* \sim \frac{\alpha_1}{1 - \rho \lambda_1} \phi_1.$$

Network best responses concentrate on agents with maximal projection onto ϕ_1 . Thus G^* converges to a rank-one kernel. $\square$

12.6. Economic Interpretation

The model predicts:

- Increasing competition raises network centralization.
- Heterogeneous risk tolerance induces assortative clustering.
- Near the critical threshold, the network collapses into a dominant core.

These results formalize how competitive benchmarking reshapes financial network topology.

13. Conclusion

This paper develops a continuous-time model of competitive portfolio choice with endogenous network formation. Agents care about relative performance and choose not only their portfolios but also the network of benchmarking relationships through which competition operates.

The key conceptual advance relative to existing graphon portfolio games is the endogenization of the interaction kernel. Rather than solving equilibrium on a fixed network, we characterize equilibrium of the network itself.

The analysis establishes four main results.

First, we prove existence of joint equilibria in portfolio strategies and graphon kernels under quadratic link costs and moderate competition intensity.

Second, we derive a spectral amplification formula showing that equilibrium risk exposures are governed by the resolvent operator $(I - \rho \mathcal{G})^{-1}$. The leading eigenvalue

of the endogenous graphon determines a critical competition threshold beyond which interior equilibria cease to exist.

Third, we show that heterogeneous risk tolerance generates assortative clustering in equilibrium networks. As competition intensifies, network centralization increases, and near the critical threshold the network collapses toward a dominant eigenmode.

Fourth, we establish that decentralized network formation is generically inefficient. Agents internalize the private benefits of amplification but neglect the systemic risk imposed on others. Only in the homogeneous benchmark case does decentralized formation coincide with the planner's allocation.

These results extend graphon-based financial games in three fundamental ways:

1. The interaction structure is no longer primitive but an equilibrium object.
2. Amplification and instability emerge endogenously from network choice.
3. Finite-agent equilibria converge to the endogenous graphon equilibrium, providing robustness of the continuum limit.

The framework provides a tractable bridge between network formation theory, competitive asset management, and systemic risk.

Several directions remain for future research. Introducing heterogeneous beliefs, dynamic network adjustment, or endogenous entry of agents would allow the study of network cycles and instability episodes. Incorporating market clearing and price impact could further connect endogenous benchmarking to asset pricing implications.

More broadly, the analysis suggests that competitive pressures not only alter trading behavior but also reshape the architecture of financial networks. Understanding this endogenous topology is essential for assessing systemic vulnerability in modern asset management systems.

References

- [1] Basak, S. and Pavlova, A. (2005). Asset prices and institutional investors. *American Economic Review* 95(5), 1728–1757.
- [2] Basak, S. and Pavlova, A. (2013). A model of financialization of commodities. *Journal of Finance* 68(4), 1511–1556.
- [3] Buffa, A., Vayanos, D., and Woolley, P. (2014). Asset management contracts and equilibrium prices. *Journal of Political Economy* 122(3), 583–613.
- [4] Frei, C. and dos Reis, G. (2011). A financial market model with interaction of investors. *Applied Mathematics and Optimization* 63, 1–27.
- [5] Espinosa, G.-E. and Touzi, N. (2015). Optimal investment under relative performance concerns. *Mathematical Finance* 25(2), 221–257.

- [6] Fu, G., Jiang, X., and Zhou, X. (2020). Mean field portfolio competition. *Finance and Stochastics* 24, 411–451.
- [7] Lasry, J.-M. and Lions, P.-L. (2007). Mean field games. *Japanese Journal of Mathematics* 2(1), 229–260.
- [8] Huang, M., Caines, P. E., and Malhamé, R. P. (2006). Large population stochastic dynamic games: closed-loop McKean–Vlasov systems and the Nash certainty equivalence principle. *Communications in Information and Systems* 6(3), 221–252.
- [9] Carmona, R. and Delarue, F. (2018). *Probabilistic Theory of Mean Field Games with Applications I–II*. Springer.
- [10] Carmona, R. and Zhu, X. (2019). Mean field games on graphons. *Stochastic Processes and their Applications* 129(11), 4675–4710.
- [11] Allen, F. and Gale, D. (2000). Financial contagion. *Journal of Political Economy* 108(1), 1–33.
- [12] Acemoglu, D., Ozdaglar, A., and Tahbaz-Salehi, A. (2015). Systemic risk and network formation. *American Economic Review* 105(2), 564–608.
- [13] Jackson, M. O. and Wolinsky, A. (1996). A strategic model of social and economic networks. *Journal of Economic Theory* 71(1), 44–74.
- [14] Tangpi, L. and Zhou, X. Y. (2023). Graphon mean field games and competitive portfolio selection. *arXiv preprint arXiv:2202.11314*.

Appendix: Detailed Derivations and Proofs

Appendix A. Gaussian–CARA certainty equivalent and the portfolio best response

This appendix section justifies the reduction from exponential utility to a mean–variance functional, and derives the best response used in the main text.

Appendix A.1. Exponential utility of a Gaussian random variable

Lemma Appendix A.1. *Let $Y \sim \mathcal{N}(m, s^2)$ and let $\gamma > 0$. Then*

$$\mathbb{E}[-e^{-\gamma Y}] = -\exp\left(-\gamma m + \frac{\gamma^2}{2} s^2\right).$$

Consequently, maximizing $\mathbb{E}[-e^{-\gamma Y}]$ over any set of controls that affect m and s^2 is equivalent to maximizing $m - \frac{\gamma}{2} s^2$.

Proof. Write $Y = m + sZ$ with $Z \sim \mathcal{N}(0, 1)$. Then

$$\mathbb{E}[e^{-\gamma Y}] = e^{-\gamma m} \mathbb{E}[e^{-\gamma s Z}].$$

The mgf of Z gives $\mathbb{E}[e^{tZ}] = e^{t^2/2}$, hence $\mathbb{E}[e^{-\gamma s Z}] = e^{\gamma^2 s^2/2}$. Multiplying yields the stated expression for $\mathbb{E}[e^{-\gamma Y}]$ and thus for $\mathbb{E}[-e^{-\gamma Y}]$. Since the map $x \mapsto -e^x$ is strictly increasing, maximizing $-\exp(-\gamma m + \frac{\gamma^2}{2}s^2)$ is equivalent to maximizing $-\gamma m + \frac{\gamma^2}{2}s^2$, which is equivalent to maximizing $m - \frac{\gamma}{2}s^2$. $\square$

Appendix A.2. Terminal relative performance is Gaussian

Assume (deterministic) linear wealth dynamics:

$$dX_t(u) = \pi_t(u)^\top \mu dt + \pi_t(u)^\top \Sigma dW_t, \quad X_0(u) = x_0(u),$$

where $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times m}$ are constant, and W is m -dimensional Brownian motion.

Integrating gives

$$X_T(u) = x_0(u) + \int_0^T \pi_t(u)^\top \mu dt + \int_0^T \pi_t(u)^\top \Sigma dW_t.$$

The stochastic integral is Gaussian with mean 0 and variance $\int_0^T \pi_t(u)^\top (\Sigma \Sigma^\top) \pi_t(u) dt$. Hence $X_T(u)$ is Gaussian for each fixed admissible strategy.

Given a measurable kernel $G(u, v)$ and a fixed population strategy profile $v \mapsto \pi(v)$, define the benchmark

$$B_T^G(u) := \int_0^1 G(u, v) X_T(v) dv,$$

and relative performance

$$Z_T(u) := X_T(u) - \rho B_T^G(u).$$

A linear combination (and L^2 -integral) of jointly Gaussian variables is Gaussian, so $Z_T(u)$ is Gaussian. Lemma [Appendix A.1](#) applies.

Appendix A.3. Derivation of the best response

Fix u and treat $B_T^G(u)$ as exogenous with respect to u 's own choice. Define

$$\bar{\pi}_t^G(u) := \int_0^1 G(u, v) \pi_t(v) dv.$$

Then the diffusion part of $Z_T(u)$ equals

$$\int_0^T (\pi_t(u) - \rho \bar{\pi}_t^G(u))^\top \Sigma dW_t.$$

Let $M := \Sigma \Sigma^\top$ (positive definite). The mean–variance criterion for $Z_T(u)$ becomes, up to constants independent of $\pi(u)$,

$$\int_0^T \left[\pi_t(u)^\top \mu - \frac{\gamma(u)}{2} (\pi_t(u) - \rho \bar{\pi}_t^G(u))^\top M (\pi_t(u) - \rho \bar{\pi}_t^G(u)) \right] dt.$$

For each t , the integrand is a strictly concave quadratic function of $\pi_t(u)$. Differentiating the integrand with respect to π gives

$$\mu - \gamma(u) M (\pi_t(u) - \rho \bar{\pi}_t^G(u)) = 0,$$

so

$$\pi_t(u) = \frac{1}{\gamma(u)} M^{-1} \mu + \rho \bar{\pi}_t^G(u).$$

Strict concavity implies uniqueness.

Appendix B. Resolvent representation of the portfolio equilibrium

Let $\mathcal{H} := L^2([0, 1]; \mathbb{R}^d)$ and define the graphon operator

$$(\mathcal{G}f)(u) := \int_0^1 G(u, v) f(v) dv.$$

Define the non-competitive (Merton-type) profile

$$a(u) := \frac{1}{\gamma(u)} M^{-1} \mu.$$

In a portfolio equilibrium given G , the best response condition yields

$$\pi = a + \rho \mathcal{G}\pi, \quad \text{in } \mathcal{H},$$

equivalently

$$(I - \rho \mathcal{G})\pi = a.$$

Whenever $(I - \rho \mathcal{G})$ is invertible on $\mathcal{H}$,

$$\pi^*(\cdot; G) = (I - \rho \mathcal{G})^{-1} a.$$

Lemma Appendix B.1 (Neumann series criterion). *If $\|\rho \mathcal{G}\|_{\text{op}} < 1$ as an operator on $\mathcal{H}$, then $(I - \rho \mathcal{G})$ is invertible and*

$$(I - \rho \mathcal{G})^{-1} = \sum_{k=0}^{\infty} (\rho \mathcal{G})^k,$$

with convergence in operator norm.

Proof. This is the standard Neumann series result for bounded linear operators: if $\|T\|_{\text{op}} < 1$ then $I - T$ is invertible with inverse $\sum_{k \geq 0} T^k$. Take $T = \rho \mathcal{G}$. $\square$

Appendix C. Functional derivatives for the network first-order condition

This section explains the first-order condition for choosing $G(u, \cdot)$ under quadratic costs, and makes explicit which quantity is being differentiated.

Appendix C.1. The network choice problem for a fixed type

Fix u . Consider kernels $g(\cdot)$ on $[0, 1]$ satisfying $g(v) \geq 0$ and $\int_0^1 g(v) dv = 1$, and interpret $g(v) = G(u, v)$. All other rows $G(s, \cdot)$ for $s \neq u$ are held fixed.

The agent's objective is

$$\mathcal{J}_u(g) = V(u; G) - \frac{\kappa}{2} \int_0^1 g(v)^2 dv,$$

where G is the full kernel with row u equal to g .

The constraint set

$$\Delta := \left\{ g \in L^2([0, 1]) : g \geq 0 \text{ a.e., } \int_0^1 g = 1 \right\}$$

is convex and closed in L^2 .

Appendix C.2. Directional derivatives

Let $h \in L^2([0, 1])$ satisfy $\int_0^1 h(v) dv = 0$. For ε small, $g + \varepsilon h$ preserves the normalization. Ignoring the inequality constraint momentarily (interior points), the directional derivative of the quadratic cost term is immediate:

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \frac{\kappa}{2} \int_0^1 (g + \varepsilon h)^2 dv = \kappa \int_0^1 g(v) h(v) dv.$$

For the value part $V(u; G)$, the dependence on g occurs through the operator $\mathcal{G}$ and hence through the equilibrium portfolio $\pi^*(\cdot; G)$. In the main text we work with the equilibrium portfolio relation

$$\pi^* = (I - \rho \mathcal{G})^{-1} a.$$

At a point where $(I - \rho \mathcal{G})$ is invertible, the map $G \mapsto \pi^*(\cdot; G)$ is differentiable in the sense of Fréchet derivatives whenever $G \mapsto \mathcal{G}$ is differentiable as a bounded operator perturbation. In particular, if $\|\rho \mathcal{G}\|_{\text{op}} < 1$, the Neumann series gives a transparent perturbation expansion:

$$\pi^*(G) = \sum_{k=0}^{\infty} \rho^k \mathcal{G}^k a.$$

A perturbation in the row u changes $\mathcal{G}$ by an operator $\delta \mathcal{G}$ that acts as

$$(\delta \mathcal{G} f)(s) = \mathbf{1}_{\{s=u\}} \int_0^1 h(v) f(v) dv,$$

in the formal “typewise” sense; in practice one works with an approximation (e.g. finitely many types / step functions) and passes to the limit.

The Euler–Lagrange condition at an interior optimum g^* is therefore the usual Lagrange multiplier statement: there exists $\lambda(u) \in \mathbb{R}$ such that

$$\langle D_g V(u; G), h \rangle_{L^2} - \kappa \int_0^1 g^*(v) h(v) dv - \lambda(u) \int_0^1 h(v) dv = 0 \quad \text{for all admissible directions } h,$$

and since $\int h = 0$, this reduces to

$$\langle D_g V(u; G), h \rangle_{L^2} = \kappa \int_0^1 g^*(v) h(v) dv, \quad \text{for all } h \text{ with } \int h = 0.$$

When $D_g V(u; G)$ admits a density $\partial V(u; G)/\partial g(v)$ in L^2 , this implies

$$\frac{\partial V(u; G)}{\partial g(v)} = \kappa g^*(v) + \lambda(u) \quad \text{a.e. in } v,$$

which is the first-order condition stated in the paper.

Appendix C.3. Including the nonnegativity constraint

If the optimum lies on the boundary where $g(v) = 0$ on a set of positive measure, the correct condition becomes a Kuhn–Tucker inequality: there exists $\lambda(u) \in \mathbb{R}$ and a multiplier $\eta(v) \geq 0$ with $\eta(v)g(v) = 0$ a.e. such that

$$\frac{\partial V(u; G)}{\partial g(v)} = \kappa g(v) + \lambda(u) - \eta(v).$$

Equivalently,

$$\frac{\partial V(u; G)}{\partial g(v)} - \lambda(u) \leq \kappa g(v), \quad \text{with equality whenever } g(v) > 0.$$

This is the correct interior/boundary characterization for the constrained problem.

Appendix D. Existence of a joint equilibrium: a clean fixed-point argument

This section provides a readable proof of the existence theorem in the main text, under the quadratic network cost and a stability condition ensuring the portfolio resolvent is well-defined.

Appendix D.1. A compact convex domain for the network

Let

$$\mathcal{G} := \left\{ G \in L^2([0, 1]^2) : G \geq 0 \text{ a.e. and } \int_0^1 G(u, v) dv = 1 \text{ for a.e. } u \right\}.$$

$\mathcal{G}$ is convex: if $G_1, G_2 \in \mathcal{G}$ and $\theta \in [0, 1]$, then $\theta G_1 + (1 - \theta)G_2 \geq 0$ and row-normalization is preserved.

To obtain compactness, one works with the weak topology in L^2 . Since L^2 is reflexive, any bounded set is relatively weakly compact.

A uniform L^2 bound can be imposed either by restricting attention to a closed ball

$$\mathcal{G}_R := \mathcal{G} \cap \{G : \|G\|_{L^2} \leq R\}$$

or by showing the best response map produces such a bound automatically under the quadratic cost. In either case, $\mathcal{G}_R$ is weakly compact and convex.

Appendix D.2. Continuity of the portfolio response

Fix ρ such that $\|\rho\mathcal{G}\|_{\text{op}} < 1$ for the relevant kernels. Then for each G , $\pi^*(\cdot; G) = (I - \rho\mathcal{G})^{-1}a$ is well-defined.

Moreover, the map $G \mapsto \mathcal{G}$ is linear and continuous from $L^2([0, 1]^2)$ into bounded operators from $L^2([0, 1])$ to itself under standard integral-operator bounds (e.g. Schur test conditions, or restricting to bounded kernels). Within such a domain, the resolvent map is continuous: small perturbations of G induce small perturbations of $\pi^*(\cdot; G)$ in L^2 .

Appendix D.3. Continuity and self-mapping of the network response

For each type u , the network best response solves a strictly convex problem:

$$\max_{g \in \Delta} \left(V(u; G) - \frac{\kappa}{2} \|g\|_{L^2}^2 \right),$$

with Δ convex and closed. Strict convexity ensures uniqueness of the optimizer when $V(u; G)$ is concave (or affine) in g on the relevant domain, or more generally when the quadratic term dominates the curvature.

Under the maintained regularity in the main text, the optimizer depends continuously on the input profile (here, the equilibrium portfolio), hence defines a continuous map $\mathcal{N}$.

Appendix D.4. Schauder fixed point

Define

$$\mathcal{T}(G) := \mathcal{N}(\pi^*(\cdot; G)).$$

$\mathcal{T}$ maps $\mathcal{G}_R$ into itself and is continuous. Since $\mathcal{G}_R$ is convex and (weakly) compact, Schauder's theorem implies the existence of $G^* \in \mathcal{G}_R$ with $\mathcal{T}(G^*) = G^*$. Define $\pi^* = \pi^*(\cdot; G^*)$. Then (G^*, π^*) is a joint equilibrium.

Appendix E. Spectral amplification under symmetry

This section justifies the spectral representation and the phase transition claim, under the additional (transparent) symmetry condition.

Appendix E.1. Compact self-adjointness

Assume $G \in L^2([0, 1]^2)$ is symmetric: $G(u, v) = G(v, u)$ a.e. Then $\mathcal{G}$ is a Hilbert–Schmidt operator on $L^2([0, 1])$, hence compact. Symmetry implies self-adjointness:

$$\langle \mathcal{G}f, g \rangle = \langle f, \mathcal{G}g \rangle, \quad \forall f, g \in L^2([0, 1]).$$

Appendix E.2. Spectral theorem and resolvent expansion

By the spectral theorem for compact self-adjoint operators, there exist orthonormal $\{\phi_k\}$ and real eigenvalues $\{\lambda_k\}$ with $\lambda_k \rightarrow 0$ such that

$$\mathcal{G}f = \sum_{k \geq 1} \lambda_k \langle f, \phi_k \rangle \phi_k.$$

Writing $a = \sum_k \alpha_k \phi_k$ and solving $(I - \rho \mathcal{G})\pi = a$ yields

$$\pi = \sum_{k \geq 1} \frac{\alpha_k}{1 - \rho \lambda_k} \phi_k,$$

whenever $1 - \rho \lambda_k \neq 0$ for all k . Let $\lambda_{\max} = \max_k \lambda_k$. Then

$$\|\pi\|_{L^2}^2 = \sum_k \frac{\alpha_k^2}{(1 - \rho \lambda_k)^2} \leq \frac{1}{(1 - \rho \lambda_{\max})^2} \sum_k \alpha_k^2 = \frac{1}{(1 - \rho \lambda_{\max})^2} \|a\|_{L^2}^2.$$

If the leading coefficient α_1 corresponding to $\lambda_{\max}$ is nonzero, then the term $\alpha_1^2/(1 - \rho \lambda_{\max})^2$ diverges as $\rho \uparrow 1/\lambda_{\max}$.

Appendix F. Planner vs decentralized formation: the externality term

This section makes explicit the exact reason the planner and decentralized conditions differ.

Appendix F.1. Planner objective

Define social welfare

$$\mathcal{W}(G) = \int_0^1 V(u; G) du - \frac{\kappa}{2} \int_0^1 \int_0^1 G(u, v)^2 dv du.$$

An interior planner optimum G^{pl} satisfies the stationarity condition

$$D\mathcal{W}(G^{\text{pl}})[H] = 0 \quad \text{for all admissible perturbations } H \text{ preserving row-normalization.}$$

Appendix F.2. Decomposition of the welfare derivative

Write

$$D\mathcal{W}(G)[H] = \int_0^1 D_G V(u; G)[H] du - \kappa \int_0^1 \int_0^1 G(u, v) H(u, v) dv du.$$

For perturbations affecting only row u (so $H(s, \cdot) \equiv 0$ for $s \neq u$), the derivative becomes

$$D\mathcal{W}(G)[H] = D_G V(u; G)[H] + \int_{s \neq u} D_G V(s; G)[H] ds - \kappa \int_0^1 G(u, v) H(u, v) dv.$$

The first term is the marginal private benefit internalized by agent u . The integral over $s \neq u$ is the externality created by changing row u : the portfolio equilibrium $\pi^*(\cdot; G)$ changes globally, affecting everyone's value.

Decentralized optimization sets

$$D_G V(u; G)[H] - \kappa \int_0^1 G(u, v) H(u, v) dv = 0$$

for all admissible row-preserving H (plus the multiplier for normalization), but does not include the externality term. Unless that externality is identically zero (typically only under strong homogeneity), planner and decentralized FOCs differ and the equilibrium is inefficient.